

Qualification Pack



Water Treatment Plant Assistant Operator

QP Code: PSC/Q0603

Version: 1.0

NSQF Level: 3.5

Water Management and Plumbing Skill Council || Unit No-606, DLF Prime Tower
Okhla Phase-1 New Delhi-110020 || email:standards@wmpsc.in



Qualification Pack

Contents

PSC/Q0603: Water Treatment Plant Assistant Operator	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
PSC/N0610: Perform Various Processes to Treat Wastewater	5
PSC/N0611: Maintain the Equipment of Water Treatment Plant	11
PSC/N0612: Perform Support Activities in the Disposal of Treated Effluent	17
PSC/N0603: Maintain Records & Reporting at Site	21
PSC/N0602: Health and Safety Practices at the treatment Plant	27
DGT/VSQ/N0101: Employability Skills (30 Hours)	31
Assessment Guidelines and Weightage	36
<i>Assessment Guidelines</i>	36
<i>Assessment Weightage</i>	37
Acronyms	38
Glossary	39

Qualification Pack

PSC/Q0603: Water Treatment Plant Assistant Operator

Brief Job Description

A Water Treatment Plant Assistant Operator is responsible for treating wastewater in water treatment plants. The individual carries out various processes for effective water treatment and effluent management in compliance with the applicable environmental regulations. The person uses different water treatment plant equipment and also carries out their basic repair and maintenance.

Personal Attributes

WTP Assistant operator requires meticulous attention to detail, strong problem-solving skills, and a methodical approach to managing wastewater treatment processes. Adaptability to changing situations, a commitment to environmental stewardship, and adherence to safety protocols are crucial. Excellent analytical abilities, a proactive attitude, and effective communication skills are essential for coordinating with teams and ensuring compliance. Additionally, the role demands a strong sense of responsibility, integrity, and a willingness to learn and implement innovative solutions for sustainable wastewater management.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSC/N0610: Perform Various Processes to Treat Wastewater](#)
2. [PSC/N0611: Maintain the Equipment of Water Treatment Plant](#)
3. [PSC/N0612: Perform Support Activities in the Disposal of Treated Effluent](#)
4. [PSC/N0603: Maintain Records & Reporting at Site](#)
5. [PSC/N0602: Health and Safety Practices at the treatment Plant](#)
6. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Plant Operations
Country	India

Qualification Pack

NSQF Level	3.5
Credits	14
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3132.0400
Minimum Educational Qualification & Experience	11th grade pass OR Previous relevant Qualification of NSQF Level (relevant Qualification of NSQF Level 2.5) with 3 Years of experience in Water Management OR Previous relevant Qualification of NSQF Level (relevant Qualification of NSQF Level 3) with 1.5 years of experience in Water Management OR 8th grade pass (relevant experience in Water Management) with 1.5 years of experience
Minimum Level of Education for Training in School	8th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	27/08/2027
NSQC Approval Date	27/08/2024
Version	1.0
Reference code on NQR	QG-3.5-WS-02940-2024-V1-WMPS
NQR Version	1.0

Remarks:

NIL

Qualification Pack

PSC/N0610: Perform Various Processes to Treat Wastewater

Description

Water Treatment Plant (WTP) operations entails managing treatment processes to purify wastewater. This involves receiving effluent, applying primary, secondary, and tertiary treatment methods, ensuring proper settling and filtration, and disinfecting the treated water. Regular equipment checks, chemical dosage, and monitoring treatment efficiency are crucial. Adjusting operations as per variations in effluent characteristics and maintaining compliance with environmental standards are key.

Scope

The scope covers the following :

- Carry out preliminary, primary and secondary water treatment
- Carry out advanced water treatment
- Treat sludge and control the odour

Elements and Performance Criteria

Carry out preliminary, primary and secondary water treatment

To be competent, the user/individual on the job must be able to:

- PC1.** Draw water in the appropriate quantity from the wastewater source, such as a river or lake.
- PC2.** Carry out preliminary treatment of water, using bar screens and mechanical rakes to remove large debris such as rags, plastics, and other solid waste.
- PC3.** Carry out primary treatment of wastewater using primary clarifiers and sedimentation tanks.
- PC4.** Skim off oils and greases floating on the top of the water using the appropriate tools and equipment.
- PC5.** Aerate the water using diffused/ mechanical aerators or oxidation ditches to break down organic matter.
- PC6.** Carry out secondary sedimentation by transferring water to secondary clarifiers where to remove activated sludge.

Carry out advanced water treatment

To be competent, the user/individual on the job must be able to:

- PC7.** Remove the remaining suspended solids and contaminants using sand filters, activated carbon filters, or membrane filters
- PC8.** Remove nutrients like nitrogen and phosphorus by following the appropriate biological or chemical processes to prevent eutrophication.
- PC9.** Eliminate the remaining pathogens using chlorine, Ultraviolet (UV) light, or ozone, as appropriate.

Treat sludge and control the odour

To be competent, the user/individual on the job must be able to:

- PC10.** Treat the sludge following the appropriate methods, such as thickening, digestion and dewatering
- PC11.** Dispose of the dewatered sludge or re-use it for appropriate purpose, e.g. soil conditioning

Qualification Pack

PC12. Break down odour-causing compounds in water using bio-filters with microorganisms

PC13. Neutralize odorous gases using chemical scrubbers

PC14. Adsorb odorous compounds activated carbon filters

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Importance and need of treating the water.

KU2. Basic terminologies like discharge, flow rate, velocity, etc

KU3. Physical characteristics of water like total solids, suspended solids, settled solids, etc.

KU4. Chemical characteristics of water like pH chloride content, nitrogen content, fats, oil & grease, sulphates, hydrogen sulphide, dissolved oxygen, etc.

KU5. Biological characteristics of water like Bod, Cod, MLSS, pH etc.

KU6. Decomposition by aerobic, anaerobic & facultative bacteria, neutralization, absorption, adsorption, oxidation/reduction, ion exchange process.

KU7. Disposal of Effluents.

KU8. Biological operations like sludge digestion, ASP, etc

KU9. Process of removal of floating, settleable organic matter and suspended solids.

KU10. Method to clean different units present in the water treatment plant.

KU11. Handling of chemicals.

KU12. Process of effluent treatment performed during treatment of chemical waste.

KU13. Different methods of handling the effluent.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. prepare checklists, report and fill out forms in local language or Hindi or English.

GS2. Understand the various color codes, as per standard electrical, mechanical and civil nomenclature.

GS3. ability to read from different sources- books screens in machines and signage.

GS4. Perform arithmetic calculations of addition, subtraction, multiplication and division processes

GS5. Read and interpret information (symbols, dimensions, terminology, dates etc.) given local language or Hindi

GS6. State information, doubts and concerns about work related matters in local language or Hindi/English.

GS7. Plan ones daily tasks to achieve maximum efficiency.

GS8. Be punctual and work as per agreed priorities

GS9. Manage distractions and maintain workplace discipline

GS10. Listen to the higher authorities carefully.

GS11. be courteous

GS12. Identify ways to increase efficiency and reduce errors.



Qualification Pack

- GS13.**
- Read and understand manuals, health and safety instructions, memos, other company documents.
 - Communicate with supervisor.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out preliminary, primary and secondary water treatment</i>	10	20	-	3
PC1. Draw water in the appropriate quantity from the wastewater source, such as a river or lake.	-	-	-	-
PC2. Carry out preliminary treatment of water, using bar screens and mechanical rakes to remove large debris such as rags, plastics, and other solid waste.	-	-	-	-
PC3. Carry out primary treatment of wastewater using primary clarifiers and sedimentation tanks.	-	-	-	-
PC4. Skim off oils and greases floating on the top of the water using the appropriate tools and equipment.	-	-	-	-
PC5. Aerate the water using diffused/ mechanical aerators or oxidation ditches to break down organic matter.	-	-	-	-
PC6. Carry out secondary sedimentation by transferring water to secondary clarifiers where to remove activated sludge.	-	-	-	-
<i>Carry out advanced water treatment</i>	12	24	-	4
PC7. Remove the remaining suspended solids and contaminants using sand filters, activated carbon filters, or membrane filters	-	-	-	-
PC8. Remove nutrients like nitrogen and phosphorus by following the appropriate biological or chemical processes to prevent eutrophication.	-	-	-	-
PC9. Eliminate the remaining pathogens using chlorine, Ultraviolet (UV) light, or ozone, as appropriate.	-	-	-	-
<i>Treat sludge and control the odour</i>	8	16	-	3
PC10. Treat the sludge following the appropriate methods, such as thickening, digestion and dewatering	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Dispose of the dewatered sludge or re-use it for appropriate purpose, e.g. soil conditioning	-	-	-	-
PC12. Break down odour-causing compounds in water using bio-filters with microorganisms	-	-	-	-
PC13. Neutralize odorous gases using chemical scrubbers	-	-	-	-
PC14. Adsorb odorous compounds activated carbon filters	-	-	-	-
NOS Total	30	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0610
NOS Name	Perform Various Processes to Treat Wastewater
Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Plant Operations
NSQF Level	3.5
Credits	3.5
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

Qualification Pack

PSC/N0611: Maintain the Equipment of Water Treatment Plant

Description

Monitoring and managing Water Treatment Plant (WTP) involves overseeing effluent quality, flow rates, and treatment processes. This includes regular testing for compliance with set standards, optimizing treatment efficiency, and maintaining equipment. Effective sludge management, data recording for compliance, and emergency preparedness are essential.

Scope

The scope covers the following :

- Carry out preventive maintenance
- Carry out corrective maintenance
- Carry out documentation and reporting

Elements and Performance Criteria

Carry out preventive maintenance

To be competent, the user/individual on the job must be able to:

- PC1.** Treat the effluent following appropriate processes, such as additional filtration or advanced oxidation to remove any contaminants
- PC2.** Coordinate appropriate tests on the effluent to ensure it meets the regulatory standards for safe disposal
- PC3.** Dispose of the effluent following the appropriate disposal methods, such as Surface Water Discharge, Groundwater Recharge, re-use and recycling, etc.
- PC4.** Ensure compliance with the environmental regulations and standards applicable to effluent disposal
- PC5.** Follow contingency plans for handling accidental spills, equipment failures, or other emergencies that could impact effluent disposal

Carry out corrective maintenance

To be competent, the user/individual on the job must be able to:

- PC6.** Conduct a root cause analysis to determine the cause for the equipment malfunction or failure
- PC7.** Isolate the faulty equipment from the system and follow lockout/tagout procedures to ensure safety
- PC8.** Repair or replace the faulty equipment components, as appropriate
- PC9.** Use the manufacturer-recommended spare parts and maintenance tools
- PC10.** Calibrate the equipment to ensure it meets the required specifications and standards
- PC11.** Conduct appropriate tests to check for the correct functioning of the repaired equipment
- PC12.** Carry out repair and maintenance within the appropriate durations as per organizational policy, to minimize potential water supply issues
- PC13.** Coordinate with the equipment manufacturer for complex repair and maintenance needs

Carry out documentation and reporting

Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC14.** Document the maintenance activities, including the nature of the fault, the steps taken for repair, parts used, and the outcome.
- PC15.** Update the equipment's maintenance history and logs to reflect the recent repair work.
- PC16.** Report maintenance activities to the plant management, outlining the issue, repair process, and recommendations to prevent future occurrences.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Different operations performed in treatment of Effluent like Screening, Grit Chamber, Flow Equalization Chamber, Detritus Tank, Aeration Tank, Skimming Tank, Primary Sedimentation Tank, Coagulation aided Sedimentation, Sludge digestion, Trickling Filters, Rotatory Biological Contractors, Membrane Bio Reactors, Moving Bed Bio Reactors, Septic Tank, Activated Sludge Process, Secondary Sedimentation Tank, Sludge Drying Beds, 3rd degree treatment.
- KU2.** Process of purifying gas in Scrubber Unit like Purifying gas evolved during sludge digestion process.
- KU3.** • Knowledge of all mechanical equipment in WTP
•
- KU4.** Method to clean different units present in water Treatment Plant.
- KU5.** Handling of chemicals.
- KU6.** Process flow of Water treatment in the plant.
- KU7.** Different methods of handling the sludge.
- KU8.** Different types of filters in WTP
- KU9.** Able to read meter and gauge reading

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare checklists, report and fill out forms in local language or Hindi or English.
- GS2.** Understand the various color codes, as per standard electrical, mechanical and civil nomenclature.
- GS3.** ability to read from different sources- books screens in machines and signage.
- GS4.** Perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS5.** Read and interpret information (symbols, dimensions, terminology, dates etc.) given local language or Hindi
- GS6.** State information, doubts and concerns about work related matters in local language or Hindi/English.
- GS7.** Plan ones daily tasks to achieve maximum efficiency.
- GS8.** Be punctual and work as per agreed priorities
- GS9.** Manage distractions and maintain workplace discipline
- GS10.** Listen to the higher authorities carefully.



Qualification Pack

- GS11.** be courteous
- GS12.** Identify ways to increase efficiency and reduce errors.
- GS13.** Read and understand manuals, health and safety instructions, memos, other company documents.
- GS14.** Communicate with supervisor.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out preventive maintenance</i>	15	30	-	4
PC1. Treat the effluent following appropriate processes, such as additional filtration or advanced oxidation to remove any contaminants	-	-	-	-
PC2. Coordinate appropriate tests on the effluent to ensure it meets the regulatory standards for safe disposal	-	-	-	-
PC3. Dispose of the effluent following the appropriate disposal methods, such as Surface Water Discharge, Groundwater Recharge, re-use and recycling, etc.	-	-	-	-
PC4. Ensure compliance with the environmental regulations and standards applicable to effluent disposal	-	-	-	-
PC5. Follow contingency plans for handling accidental spills, equipment failures, or other emergencies that could impact effluent disposal	-	-	-	-
<i>Carry out corrective maintenance</i>	10	20	-	3
PC6. Conduct a root cause analysis to determine the cause for the equipment malfunction or failure	-	-	-	-
PC7. Isolate the faulty equipment from the system and follow lockout/tagout procedures to ensure safety	-	-	-	-
PC8. Repair or replace the faulty equipment components, as appropriate	-	-	-	-
PC9. Use the manufacturer-recommended spare parts and maintenance tools	-	-	-	-
PC10. Calibrate the equipment to ensure it meets the required specifications and standards	-	-	-	-
PC11. Conduct appropriate tests to check for the correct functioning of the repaired equipment	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Carry out repair and maintenance within the appropriate durations as per organizational policy, to minimize potential water supply issues	-	-	-	-
PC13. Coordinate with the equipment manufacturer for complex repair and maintenance needs	-	-	-	-
<i>Carry out documentation and reporting</i>	5	10	-	3
PC14. Document the maintenance activities, including the nature of the fault, the steps taken for repair, parts used, and the outcome.	-	-	-	-
PC15. Update the equipment's maintenance history and logs to reflect the recent repair work.	-	-	-	-
PC16. Report maintenance activities to the plant management, outlining the issue, repair process, and recommendations to prevent future occurrences.	-	-	-	-
NOS Total	30	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0611
NOS Name	Maintain the Equipment of Water Treatment Plant
Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Plant Operations, Plumbing Plant Operations
NSQF Level	3.5
Credits	3.5
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

Qualification Pack

PSC/N0612: Perform Support Activities in the Disposal of Treated Effluent

Description

Effluent waste disposal involves the safe discharge of effluent into water bodies, reuse for non-potable purposes like irrigation, or land application, Compliance with regulations, continuous monitoring, and measures to minimize environmental impact are crucial for efficient and sustainable effluent waste disposal practices.

Scope

The scope covers the following :

- Treat and disposal of effluent
- monitor discharge points and maintain discharge infrastructure

Elements and Performance Criteria

Treat and dispose of effluents

To be competent, the user/individual on the job must be able to:

- PC1.** Treat the effluent following appropriate processes, such as additional filtration or advanced oxidation to remove any contaminants
- PC2.** Coordinate appropriate tests on the effluent to ensure it meets the regulatory standards for safe disposal
- PC3.** Dispose of the effluent following the appropriate disposal methods, such as Surface Water Discharge, Groundwater Recharge, re-use and recycling, etc.
- PC4.** Ensure compliance with the environmental regulations and standards applicable to effluent disposal
- PC5.** Follow contingency plans for handling accidental spills, equipment failures, or other emergencies that could impact effluent disposal
- PC6.** Maintain appropriate records concerning the effluent quality, discharge and disposal

Monitor discharge points and maintain discharge infrastructure

To be competent, the user/individual on the job must be able to:

- PC7.** Monitor the discharge points to track effluent quality and ensure compliance with environmental standards
- PC8.** Maintain the discharge infrastructure, including pipes and channels to prevent leaks, blockages, and equipment failures
- PC9.** Maintain appropriate records concerning the monitoring of discharge points and maintenance of discharge infrastructure

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** importance of monitoring effluent quality regularly and complying with regulatory standards is essential

Qualification Pack

- KU2.** Sampling, testing, and analyzing effluent characteristics to ensure that treatment processes.
- KU3.** guidelines for effluent treatment.
- KU4.** Knowledge of different treatment methods.
- KU5.** advantages and limitations based on the effluent's composition and desired outcomes.
- KU6.** various sources of effluents (such as industrial processes, agricultural runoff, or municipal sewage) and their composition.
- KU7.** impact of effluent discharge on the environment is crucial for effective waste management.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare checklists, report and fill out forms in local language or Hindi or English.
- GS2.** Understand the various color codes, as per standard electrical, mechanical and civil nomenclature.
- GS3.** ability to read from different sources- books screens in machines and signage.
- GS4.** Perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS5.** Read and interpret information (symbols, dimensions, terminology, dates etc.) given local language or Hindi
- GS6.** State information, doubts and concerns about work related matters in local language or Hindi/English.
- GS7.** Plan ones dailly tasks to achieve maximum efficiency.
- GS8.** Be punctual and work as per agreed priorities
- GS9.** Manage distractions and maintain workplace discipline
- GS10.** Listen to the higher authorities carefully.
- GS11.** be courteous
- GS12.** Identify ways to increase efficiency and reduce errors.
- GS13.** Read and understand manuals, health and safety instructions, memos, other company documents.
- GS14.** Communicate with supervisor.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Treat and dispose of effluents</i>	15	30	-	5
PC1. Treat the effluent following appropriate processes, such as additional filtration or advanced oxidation to remove any contaminants	-	-	-	-
PC2. Coordinate appropriate tests on the effluent to ensure it meets the regulatory standards for safe disposal	-	-	-	-
PC3. Dispose of the effluent following the appropriate disposal methods, such as Surface Water Discharge, Groundwater Recharge, re-use and recycling, etc.	-	-	-	-
PC4. Ensure compliance with the environmental regulations and standards applicable to effluent disposal	-	-	-	-
PC5. Follow contingency plans for handling accidental spills, equipment failures, or other emergencies that could impact effluent disposal	-	-	-	-
PC6. Maintain appropriate records concerning the effluent quality, discharge and disposal	-	-	-	-
<i>Monitor discharge points and maintain discharge infrastructure</i>	15	30	-	5
PC7. Monitor the discharge points to track effluent quality and ensure compliance with environmental standards	-	-	-	-
PC8. Maintain the discharge infrastructure, including pipes and channels to prevent leaks, blockages, and equipment failures	-	-	-	-
PC9. Maintain appropriate records concerning the monitoring of discharge points and maintenance of discharge infrastructure	-	-	-	-
NOS Total	30	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0612
NOS Name	Perform Support Activities in the Disposal of Treated Effluent
Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Plant Operations
NSQF Level	3.5
Credits	3.5
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

Qualification Pack

PSC/N0603: Maintain Records & Reporting at Site

Description

This unit focuses on the importance of maintaining accurate records and effectively reporting activities at a sewage treatment plant. It emphasizes the significance of documentation in ensuring compliance with regulations, monitoring plant performance, and facilitating continuous improvement. Trainees will learn how to record data, prepare reports, and communicate relevant information to stakeholders.

Scope

The scope covers the following :

- Understand how to maintain records and data
- Reporting to the higher officials
- Respond to higher officials

Elements and Performance Criteria

Understand how to maintain records and data

To be competent, the user/individual on the job must be able to:

- PC1.** able to read plan & specifications for different types of modifications in sewage treatment plant.
- PC2.** able to document data on pathogen reduction.
- PC3.** able to record the data on sludge accumulated and sludge treated.
- PC4.** able to maintain daily operational log.
- PC5.** perform routine operational duties, flow meter readings, sample information with corresponding analytical data, etc.
- PC6.** able to identify unusual conditions at the site
- PC7.** identify and report accident equipment hazard.
- PC8.** perform preventive maintenance

Reporting to the higher officials

To be competent, the user/individual on the job must be able to:

- PC9.** reporting to the supervisor if any kind of problem occurs
- PC10.** identify spare parts inventory when in need
- PC11.** identify specifications on equipment from supplier
- PC12.** equipment inventory

Respond to higher officials

To be competent, the user/individual on the job must be able to:

- PC13.** able to responding promptly to higher officials' inquiries or requests demonstrates a sense of urgency and professionalism.
- PC14.** able to provide clear and accurate information in responses to higher officials ensures effective communication and minimizes misunderstandings.

Qualification Pack

- PC15.** able to ensure that responses align with relevant policies, procedures, and regulations demonstrates a commitment to adhering to organizational guidelines.
- PC16.** able to maintain respectful and courteous tone in responses conveys professionalism and enhances the organization's reputation.
- PC17.** identify potential solutions or recommendations in responses demonstrates proactive thinking and a commitment to finding effective resolutions.
- PC18.** able to follow up on requests or inquiries to higher officials and providing updates or additional information as needed helps to maintain effective communication and ensure that all concerns are addressed satisfactorily.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand local, state, and national regulations related to sewage treatment, including record-keeping and reporting requirements mandated by environmental agencies
- KU2.** Familiarity with different types of records, such as daily log sheets, maintenance logs, chemical usage reports, effluent quality records, energy consumption, and equipment performance logs.
- KU3.** Knowledge of how to sample influent and effluent water, record results from tests (e.g., pH, BOD, COD, TSS), and ensure these meet environment
- KU4.** Knowledge of keeping detailed logs of routine and emergency maintenance activities on equipment (e.g., pumps, aerators, clarifiers) to track equipment lifespan, repairs, and prevent failures.
- KU5.** Understand protocols for reporting spills, overflows, equipment malfunctions, or non-compliance events to regulatory authorities as required.
- KU6.** Knowledge of SOPs for different tasks within the sewage treatment plant, and the importance of maintaining updated records that align with these procedures.
- KU7.** Documenting the scheduled preventive maintenance tasks and keeping track of their completion to minimize downtime and optimize plant performance.
- KU8.** Knowledge to record data related to odor, noise, and emissions (if applicable) to ensure the plants environmental impact is minimized and within regulatory limits.
- KU9.** Understand the use of digital systems for data logging, report generation, and data retrieval. These systems may include SCADA (Supervisory Control and Data Acquisition) or other monitoring tools.
- KU10.** Generate performance reports for internal review, management oversight, and submission to regulatory bodies.
- KU11.** Knowledge of proper archiving of records, both in digital and hard copy formats, in accordance with legal retention policies, and ensuring easy retrieval during audits.
- KU12.** Knowledge of documentation procedures during emergencies (e.g., system failures, flooding) and having templates for swift reporting in urgent situations.

Generic Skills (GS)

User/individual on the job needs to know how to:



Qualification Pack

- GS1.** prepare checklists, reports and fill out forms in local language or hindi/english
- GS2.** ability to read from different sources- books screens in machines and signage.
- GS3.** perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS4.** read and interpret information (symbols, dimensions, terminology, dates etc.) given in local language or hindi
- GS5.** state information, doubts and concerns about work related matters in local language or hindi/english
- GS6.** listen to the higher authorities carefully
- GS7.** communicate with supervisor.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understand how to maintain records and data</i>	10	20	-	3
PC1. able to read plan & specifications for different types of modifications in sewage treatment plant.	-	-	-	-
PC2. able to document data on pathogen reduction.	-	-	-	-
PC3. able to record the data on sludge accumulated and sludge treated.	-	-	-	-
PC4. able to maintain daily operational log.	-	-	-	-
PC5. perform routine operational duties, flow meter readings, sample information with corresponding analytical data, etc.	-	-	-	-
PC6. able to identify unusual conditions at the site	-	-	-	-
PC7. identify and report accident equipment hazard.	-	-	-	-
PC8. perform preventive maintenance	-	-	-	-
<i>Reporting to the higher officials</i>	10	20	-	4
PC9. reporting to the supervisor if any kind of problem occurs	-	-	-	-
PC10. identify spare parts inventory when in need	-	-	-	-
PC11. identify specifications on equipment from supplier	-	-	-	-
PC12. equipment inventory	-	-	-	-
<i>Respond to higher officials</i>	10	20	-	3
PC13. able to responding promptly to higher officials' inquiries or requests demonstrates a sense of urgency and professionalism.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. able to provide clear and accurate information in responses to higher officials ensures effective communication and minimizes misunderstandings.	-	-	-	-
PC15. able to ensure that responses align with relevant policies, procedures, and regulations demonstrates a commitment to adhering to organizational guidelines.	-	-	-	-
PC16. able to maintain respectful and courteous tone in responses conveys professionalism and enhances the organization's reputation.	-	-	-	-
PC17. identify potential solutions or recommendations in responses demonstrates proactive thinking and a commitment to finding effective resolutions.	-	-	-	-
PC18. able to follow up on requests or inquiries to higher officials and providing updates or additional information as needed helps to maintain effective communication and ensure that all concerns are addressed satisfactorily.	-	-	-	-
NOS Total	30	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0603
NOS Name	Maintain Records & Reporting at Site
Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Plant Operations
NSQF Level	4
Credits	0.5
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

Qualification Pack

PSC/N0602: Health and Safety Practices at the treatment Plant

Description

this unit is about the health-related issues at stp and safety practices which workers need to be taken care off. health and safety practices plays a crucial role to ensure the well-being and protection of employees, visitors, and the environment. these practices aim to minimize the risks associated with handling and treating wastewater, hazardous chemicals, and potential biological contaminants.

Scope

The scope covers the following :

- Maintaining health at treatment plant
- safe work practises at the Treatment facilities

Elements and Performance Criteria

maintaining health at treatment plant

To be competent, the user/individual on the job must be able to:

- PC1.** identify the hazards associated with different tasks in a wastewater treatment plant.
- PC2.** able to understand and acquire know-how of safety measures to be taken when in project site.
- PC3.** identify general risks, including working at height and hazards associated with different tasks in a sewage treatment plant.
- PC4.** understand and acquire know-how of safety measures when handling mechanical parts in statutory or moving position.
- PC5.** perform the changing of chlorine cylinders safely.
- PC6.** perform the automatic and manual functions at scrubber unit safely as per instructions.

safe work practices at treatment facilities

To be competent, the user/individual on the job must be able to:

- PC7.** inform the higher authorities of the accident or hazard that has occurred.
- PC8.** select the right personal protective equipment for different tasks in a waste treatment plant.
- PC9.** ensure safe storage of corrosive substances.
- PC10.** follow recommended personal hygiene and sanitation practices, for example, washing/sanitizing hands, covering face with
- PC11.** able to clean and disinfect work area, materials/supplies, equipment etc.. before and after use.
- PC12.** able to identify hygiene and sanitation issues.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** possible causes of risk, hazard or accident in the workplace

Qualification Pack

- KU2.** location of all the general health and safety equipment in the workplace
- KU3.** work practices and precautions to control and prevent risks, hazards and accidents
- KU4.** meaning of hazards and risks
- KU5.** do and donts in operation of waste treatment plant
- KU6.** practise of safe storage of corrosive substances.
- KU7.** able to detect chlorine leakage and control, scrubber unit gas leakage and control

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare checklists, reports and fill out forms in local language or hindi/english
- GS2.** understand the various colour codes, as per standard electrical, mechanical and civil nomenclature.
- GS3.** perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS4.** state information, doubts and concerns about work related matters in local language or hindi/english
- GS5.** plan one's daily tasks to achieve maximum efficiency
- GS6.** be punctual and work as per agreed priorities
- GS7.** manage distractions and maintain workplace discipline
- GS8.** listen to the higher authorities carefully
- GS9.** be courteous
- GS10.** read and understand manuals, health and safety instructions, memos, other company documents

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>maintaining health at treatment plant</i>	15	30	-	5
PC1. identify the hazards associated with different tasks in a wastewater treatment plant.	-	-	-	-
PC2. able to understand and acquire know-how of safety measures to be taken when in project site.	-	-	-	-
PC3. identify general risks, including working at height and hazards associated with different tasks in a sewage treatment plant.	-	-	-	-
PC4. understand and acquire know-how of safety measures when handling mechanical parts in statutory or moving position.	-	-	-	-
PC5. perform the changing of chlorine cylinders safely.	-	-	-	-
PC6. perform the automatic and manual functions at scrubber unit safely as per instructions.	-	-	-	-
<i>safe work practices at treatment facilities</i>	15	30	-	5
PC7. inform the higher authorities of the accident or hazard that has occurred.	-	-	-	-
PC8. select the right personal protective equipment for different tasks in a waste treatment plant.	-	-	-	-
PC9. ensure safe storage of corrosive substances.	-	-	-	-
PC10. follow recommended personal hygiene and sanitation practices, for example, washing/sanitizing hands, covering face with	-	-	-	-
PC11. able to clean and disinfect work area, materials/supplies, equipment etc.. before and after use.	-	-	-	-
PC12. able to identify hygiene and sanitation issues.	-	-	-	-
NOS Total	30	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0602
NOS Name	Health and Safety Practices at the treatment Plant
Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Plant Operations
NSQF Level	4
Credits	0.5
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 70% of %

Qualification Pack

aggregate marks to successfully clear the assessment.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSC/N0610.Perform Various Processes to Treat Wastewater	30	60	0	10	100	35
PSC/N0611.Maintain the Equipment of Water Treatment Plant	30	60	0	10	100	15
PSC/N0612.Perform Support Activities in the Disposal of Treated Effluent	30	60	0	10	100	25
PSC/N0603.Maintain Records & Reporting at Site	30	60	0	10	100	10
PSC/N0602.Health and Safety Practices at the treatment Plant	30	60	0	10	100	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
Total	170	330	-	50	550	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.